Quiz 4 - Discrete Mathematics (CSE121)

Date: 15-11-2025 **Maximum marks**: 30

Instructions. Please read the following points carefully.

- Mention your names, roll numbers, room number, and the part number of the quiz in your answer sheets. There will be a penalty associated with missing any of these details.
- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- If you are found copying, the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or TAs.
- This exam is designed for 50 minutes.

1 Problem Set

Question 1.1 (10 marks)**.** *Let A be the set of all functions from $\mathbb{N}$ to $\mathbb{N}$. Prove that A is uncountable. You can directly use the results taught in the lectures.*

Solution: *To show A is uncountable it is sufficient to show there exists a subset of A which is uncountable. Let us take the set A' which is the set of all functions from $\mathbb{N}$ to $\{0,1\}$. We are done if we can show $A' \subseteq A$ and A' is uncountable.*

(1) A' is a subset of A.

Let $f \in A'$. By definition $f : \mathbb{N} \to \{0,1\}$. Since $\{0,1\} \subseteq \mathbb{N}$, the same map f can be regarded as a function $f : \mathbb{N} \to \mathbb{N}$. Therefore $f \in A$. As this holds for every $f \in A'$, we have

$$A' \subseteq A.$$

This completes the argument that A' is a subset of A.

(2) A' is uncountable.

Let us first prove that A' is infinite.

To prove this, it is sufficient to show that there exists a subset S of A' which is infinite.

Let

$$S = \{f_1, f_2, f_3, \ldots\}$$

where each $f_i : \mathbb{N} \to \{0,1\}$ is defined as

$$f_i(x) = \begin{cases} 1, & \textit{when } x = i, \\ 0, & \textit{when } x \neq i. \end{cases}$$

As for any $f_i \in S$, we have $f_i \in A'$, therefore $S \subseteq A'$.

Now, we will show that for $i \neq j$, we have $f_i \neq f_j$. Hence, S is infinite.

Let us assume, for contradiction, that there exist $i, j \in \mathbb{N}$ such that $i \neq j$ and $f_i = f_j$. Then

$$f_i(x) = f_j(x) \quad \forall x \in \mathbb{N}.$$

Hence $f_i(i) = f_j(i)$. But $f_i(i) = 1$ while $f_j(i) = 0$ for $i \neq j$, a contradiction. Therefore $f_i \neq f_j$ whenever $i \neq j$, so S is infinite.

As $S \subseteq A'$ and S is countably infinite, A' is infinite. *(1)*

Now we will prove that A', which is the set of all functions from $\mathbb{N}$ to $\{0, 1\}$, is uncountable.

Assume, for contradiction, that A' is countable. From (1), as A' is infinite, we can assume that A' is countably infinite. Thus there exists a bijection

$$B : \mathbb{N} \to A'.$$

Write, for each $i \in \mathbb{N}$,

$$B(i) = f_i,$$

where each $f_i \in A'$.

Define a function $f \in A'$ by

$$f(i) = 1 - f_i(i) \quad \text{for every } i \in \mathbb{N}.$$

We claim that f is not equal to any f_i.

Suppose, for contradiction, that $f = f_j$ for some j. Then

$$f(j) = f_j(j).$$

But by definition $f(j) = 1 - f_j(j)$. Hence

$$f_j(j) = 1 - f_j(j),$$

which implies $2f_j(j) = 1$. This is impossible since $f_j(j)$ is either 0 or 1. Contradiction.

Therefore f is not in the image of B, so B is not surjective, contradicting the assumption that B is a bijection. Thus A' is uncountable.

Since $A' \subseteq A$ and A' is uncountable, it follows that A is uncountable.

Hence proved that A is uncountable. $\square$

Rubrics (10 Marks)

- *(2 mark) Showing that A' is infinite by constructing an infinite subset $S \subseteq A'$.*
- *(1 mark) Writing the correct contradiction step: Assuming A' is countably infinite.*
- *(7 marks) Providing a correct Cantor diagonalisation argument to prove that A' is uncountable.*

Question 1.2. 1. *(6 marks). Let p be a prime number. For every $a, b \in \mathbb{N}$ such that $a \neq 0, b \neq 0$, prove that p divides ab if and only if p either divides a or p divides b. You should prove it from scratch and are not allowed to directly use anything done in the lectures.*

Solution: *Given in the lecture notes.*

Rubric:

- *1 mark for stating the claim clearly and sets up $p|ab$ using the definition of divisibility.*
- *1 mark for correctly argues gcd argument.*
- *1 marks for case analysis.*
- *2 mark for Euclid's Lemma.*
- *1 mark for backward direction.*

2. *(4 marks). Prove that for every $n \in \mathbb{N}, n > 1$,*

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}.$$

Solution: *We want to prove that for every integer $n > 1$,*

$$1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}.$$

Let

$$S_n = 1 + \frac{1}{4} + \frac{1}{9} + \cdots + \frac{1}{n^2} < 2 - \frac{1}{n}.$$

Base Case: *for $n = 2$, it is true. (Check by putting the values in the both sides of the equation.)*

Induction Hypothesis: *Assuming the statement is true for n*

Induction Step: *Do the analysis and calculation for $n + 1$*

Rubric:

- *1 mark for correct base case.*
- *1 mark for correct induction hypothesis.*
- *2 mark for correct induction step.*

Question 1.3. *1. (5 marks). A licence number is formed using 26 upper case Roman letters and numbers between 0 to 9. A licence number has the following structure:*

- *Total number of characters in every licence number is between 5 to 8.*
- *The first character is a non-zero even number, followed by a letter bigger than the letter C, and the the last two characters are distinct numbers.*
- *The remaining characters can be any thing from the letters and numbers mentioned above.*

Some examples of such numbers are 2DRE67, 4M789, etc.

How many such licence numbers are there? You can use the combinatorial rules taught in the class.

Solution:

Step 1: Choices for the fixed parts

- *First character: non-zero even digit : $\{2, 4, 6, 8\} = 4$ ways.*

- *Second character: letter > 'C' =* $\{D, \ldots, Z\} : 26 - 3 = 23$ *ways.*
- *Last two characters: distinct digits (0–9):* $10 \times 9 = 90$ *ways. (Product Rule)*

Step 2: Middle characters

- *Total length* $n \in \{5, 6, 7, 8\}$.
- *Number of middle characters* $= n - 4$ *(positions 3 to* $n - 2$ *inclusive).*
- *Each middle character: 36 choices (26 letters + 10 digits).*
- *So for each* n : *Count*$(n) = 4 \times 23 \times 36^{n-4} \times 90$ *(Product Rule)*

Applying sum rule

Since the licence length can be 5, 6, 7, or 8, we add the counts for each case:

$$= 4 \times 23 \times 90 \times \sum_{m=1}^{4} 36^m = 14304561120.$$

Rubric:

- *1 mark for each correct decision about the choices of characters. There are four decision.*
- *1 mark for correct final answer. Not necessary to do the whole calculation as it is a big number.*
- *-1 mark for informal answer like missing the necessary rules such as procut or sum rule.*

2. *(5 marks). Let* A, B *be finite sets. Prove that* $A \cup B, A \cap B, pow(A), A \times B$ *and* $A - B$ *are also finite sets, where* $pow(A)$ *is the power set of* A.

Solution.

Let A *and* B *be finite sets. Write* $|A| = m$ *and* $|B| = n$, *where* $m, n \in \mathbb{N}$.

(1) $A \cup B$ *is finite. Using inclusion–exclusion,*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Since $|A| = m$ *and* $|B| = n$ *are finite and* $|A \cap B| \geq 0$, *we get* $|A \cup B| \leq m + n$, *so* $A \cup B$ *is finite.*

(2) $A \cap B$ *is finite.* $A \cap B \subseteq A$. *Any subset of a finite set is finite, so* $A \cap B$ *is finite.*

(3) $\mathrm{pow}(A)$ *is finite. If* $|A| = m$, *enumerate* $A = \{a_1, \ldots, a_m\}$. *Each subset* $S \subseteq A$ *corresponds to an* m*-tuple* $(b_1, \ldots, b_m) \in \{0, 1\}^m$ *with* $b_i = 1$ *iff* $a_i \in S$. *There are* 2^m *such tuples, so* $|\mathrm{pow}(A)| = 2^m$, *hence finite.*

(4) $A \times B$ *is finite. If* $A = \{a_1, \ldots, a_m\}$ *and* $B = \{b_1, \ldots, b_n\}$ *then elements of* $A \times B$ *are ordered pairs* (a_i, b_j) *with* $1 \leq i \leq m$, $1 \leq j \leq n$. *By the product rule* $|A \times B| = m \cdot n$, *so* $A \times B$ *is finite.*

(5) $A - B$ *is finite.* $A - B \subseteq A$, *so as a subset of a finite set it is finite.*

Rubric:

For each of the five statements (1)–(5) award up to 1 mark as follows:

- *0.5 mark — Correct statement/result (i.e., correctly states that the set in question is finite and, if numeric, gives the correct count or bound).*
- *0.5 mark — Correct justification (a clear, valid argument: subset argument, inclusion–exclusion, counting/bijection to* $\{0,1\}^m$*, or product rule as appropriate).*